

Exploring Human Factors in Smart Manufacturing (Industry 4.0): A Literature Review

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The rapid development of technologies that are digital in manufacturing has driven a shift toward Industry 4.0, where human operators collaborate with automation, robotics, and data-driven systems. Despite these technologies supplying greater adaptability and productivity, their impact on human performance and well-being are currently being evaluated. To understand how ergonomics, cognitive workload, trust in automation, interface design, and training affect the effectiveness of smart manufacturing systems, this study reviews 10 peer-reviewed research papers that were published between 2015 and 2025. Articles were retrieved from IEEE Xplore, ScienceDirect, and Web of Knowledge and analyzed according to their human-factor themes and practical insights. Increasing system complexity, the findings indicate, can pose challenges related to increased mental demands, ambiguous feedback, and inconsistent confidence in the decisions of automated systems. The review emphasizes the increasing relevance of human-machine collaboration, participatory design, and continuous training. Practical implications are addressed with a view to endorse the development of safe, human-centered Industry 4.0 environments.

INTRODUCTION

In the manufacturing industry, continuous improvement is often implemented to increase efficiency. One example of this is using Industry 4.0 in smart manufacturing as it is known to resolve some of the bottlenecks that are present in production lines. Industry 4.0 has many improvements to increase the overall efficiency. For example, they have AI scheduling, robotics, and semi/full automation. These are used to reshape the current human roles in production. Though these new technologies improve efficiency, they also bring forth a few human factors challenges such as operator stress, trust and reliance on automation, user interface ergonomics, and possible skill gaps/training requirements. This paper reviews recent literature on the collaborations between human and automation techniques in the manufacturing industry.

Human factors can be responsible for up to 70-80% of industrial accidents. There is also an impact on productivity and employees' sense of achievement (Virmani & Salve, 2023). These are the major stumbling blocks to successful implementation of Industry 4.0. With the increasing automation of routine physical tasks, human operators are called upon to fulfill roles which demand high cognitive engagement such as monitoring, problem-solving, and decision-making under circumstances involving a lot of uncertainty. This transformation requires an examination of cognitive ergonomics, appropriate training, and system design that supports human effectiveness.

Since AI is being used more in various industries, it would be helpful to also implement it in the manufacturing industry. Researching industry 4.0 is important because it will allow us to learn more about the current manufacturing industry and how it can be improved in the future. Through this research we can learn what strategies are able to efficiently support safe and ethical manufacturing processes.

This literature review merges research about the confluence of human factors engineering and Industry 4.0 technologies in production. The focus will be to look into (1) the main human factors challenges that spring from introducing

Industry 4.0 technologies; (2) design strategies and countermeasures that have proved successful in covering these challenges, and (3) the gaps in our present knowledge and practical applications. This paper describes the methodology used to identify and analyze relevant literature, presents organized findings, discusses implications and limitations of current research, and concludes with recommendations for practice and future research directions.

METHOD

In November 2025, a literature review was conducted that was focused primarily on scholarly articles regarding human factors within the context of Industry 4.0 or industrial automation. The search strategy was deployed throughout the major databases of scholarly literature, including IEEE Xplore, ScienceDirect, and Web of Knowledge, to ensure the broadest coverage of relevant literature. The search was conducted by incorporating various phrases such as 'human factors', 'smart manufacturing', 'Industry 4.0', 'ergonomics' and 'human-robot cooperation', 'automation trust', or 'operator 4.0'. A time range of the years 2015 to 2025 was used so that the information would remain relevant to the current industry.

The papers that were picked out during this review addressed human factors and ergonomics in automated manufacturing contexts, provided data or theoretical frameworks about human-automation interaction, and focused on industrial or production environments. The analysis framework allowed for each selected study to be reviewed according to the following four criteria: study type, technology studied, human factors addressed, and the key findings. There was also an emphasis on finding articles that included practical implications of the manufacturing system design.

RESULTS

The literature review pointed out four main themes: ergonomic considerations, trust and human-robot collaboration, interface design, and human centric design. Refer to Table 1 for

a brief overview of the research papers and their contributions to the literature review.

Table 1. Overview of Research Papers

Authors	Year	Primary Focus	Key Finding
Cao et al.	2016	Ergonomic models	Developed data-driven evaluation methods
Castagnoli et al.	2024	Human-centric manufacturing	Identified need for holistic integration
Gladysz et al.	2023	Operator 4.0 typology	Defined framework for human roles in Industry 4.0
Kamble et al.	2018	Implementation barriers	Highlighted training and resistance issues
Khafaga et al.	2023	Augmented reality systems	Showed AR reduces cognitive load and errors
Longo et al.	2020	Ethical technology design	Emphasized values beyond productivity
Mattsson et al.	2020	Cognitive models	Analyzed trust development mechanisms
Neumann et al.	2021	Systems framework	Proposed integrated methodology for human-system design
Reiman et al.	2021	Ergonomics research agenda	Identified cognitive load as emerging concern
Virmani & Salve	2023	HFE and operational excellence	Demonstrated HFE mediates Industry 4.0 success

There is a bigger emphasis on human-machine collaboration, rather than trying to implement full automation. Ergonomic improvements such as tactiles are being implemented. For example, wearable sensors and smart interfaces. It is shown that manufacturing companies that have a strong training and feedback system tend to show a higher acceptance of new technologies being implemented. More machines/automation means that human operators will have to do more multitasking and be overloaded with more information. It can be hard to adjust to new machines let alone automation, which is still in its earlier stages. There are also issues that come with mistrust and over/under reliance on the AI-driven decision systems.

The concept of Industry 4.0 signifies a significant shift in the manufacturing industry by combining automation, cyber-physical systems, and artificial intelligence. But regardless of these technologies have increased production and efficiency,

they have also created new issues with decision-making, human roles, and system usability. Another aspect that was emphasized was the importance of humans even in automated environments by describing the Operator 4.0 as a competent and flexible worker who works with intelligent machines (Gladysz et al., 2023). Nevertheless, with technology replacing physical labor and tedious tasks, humans are increasingly assuming demanding tasks of monitoring, supervision, and problem-solving. This trend focuses renewed attention on ergonomics, cognitive workload, and human-centered system design (Neumann et al., 2021; Virmani & Salve, 2023). The most recent research further emphasized the critical nature of establishing an appropriate balance between advances in technology and employee productivity and mental health, suggesting that human factors engineering (HFE) is a fundamental rather than elective part of healthy manufacturing processes.

Ergonomics

Automation has minimized the physical stress placed on employees, yet it has also brought forth intellectual demands that are frequently forgotten. According to Neumann et al. (2021), multitasking, constant information flow, and system oversight cause operators in smart factories to experience stress. By combining organizational, cognitive, and physical ergonomics, Reiman et al. (2021) built on this and suggested that Industry 4.0 work systems need adaptable designs that facilitate feedback clarity and attention management. Data-driven evaluation methods that use wearable sensors to track posture and mental strain are part of recent ergonomic models created (Cao et al., 2016). These findings suggest that cognitive strain and attentional weakness are emerging as the main ergonomic concerns, even though robots could eventually take over physical labor. In the modern period of digital production, creating intelligent technologies to assist employees in processing information effectively and avoiding overload has emerged as a major human-factors issue.

Trust and Human-Robot Collaboration

As human-robot collaboration is growing in popularity, trust has become a key component of success. Strong HFE practices, such as safety training, ergonomic interface design, and clear communication protocols, have been empirically shown by Virmani and Salve (2023) to improve performance and employee trust in automation. To understand how human confidence in robots develops, cognitive models have been created (Mattsson et al., 2020). They cautioned that although lack of trust may result in inadequate utilization of automation, dependence can lead to complacency. According to Longo et al. (2020), ethical design, openness, and consistent robot behavior promote calibrated trust in which operators are aware of the advantages and disadvantages of automated systems. When combined, these studies demonstrate that trust is a quantifiable result of design, transparency, and system behavior rather than an ethereal feeling. Even the best technologies may not be accepted by users if they mistrust, under reliance or overreliance, the machines.

Usability

An understandable interface design is essential to make sure human operators can engage with automated systems. Khafaga et al. (2023) suggested that participatory design, where operators are directly involved in refining dashboards testing and, to improve usability, and minimize confusion. Visual displays that are purely structured as reported in various research, increase cognitive workload and lead to decision fatigue (Reiman et al., 2021). Kamble et al. (2018) explored the use of augmented reality (AR) assistance systems that overlay task instructions and status information in the operator's field of view, reducing cognitive load, and improving accuracy. Through modifying information to each operator's task, these augmented reality interfaces increase engagement and speed up reaction times. Although these results imply that human-centered design principles emphasize clarity, simplicity, and interface comfort are more important for system usability than software capabilities.

Human-Centric Direction

In industry 4.0 to sustain innovation, staff training and adaptation were crucial. According to Kamble et al. (2018), in contrast to change and inadequate training remain significant barriers to digital transformation, specifically in familiar manufacturing settings. A recent comprehensive evaluation of human-centric manufacturing Castagnoli et al. (2024) found that for full adoption, organizational, technical, and human factors must be included in future studies. A quick shift to industry 5.0 should encourage human creativity, ethics, and empathy along with productivity, Longo et al. (2020). This shows a raising awareness that collaborating alongside machines increases productivity if they can communicate effectively. As a result, it's critical to develop a workforce with a human-centric design while taking participatory ergonomics, ethical design, and continuous training into consideration

DISCUSSION

Through the results, it becomes apparent that human factors play a huge role in the success of Industry 4.0. Even though automation allows for increased productivity, it also introduces challenges related to ergonomics, trust dynamics, interface usability, and adaptation. Across this literature review there are various insights that are brought forth while elaborating about using automation as help for the human operators rather than replacing them completely. The Operator 4.0 concept talks more about this, positioning humans as the partners of automated machines and training them how to use it properly (Gładysz et al., 2023). This is better than the alternative where the workers only handle the tasks that are not yet automated and, in the end, get replaced by full automation.

Next, interface design also plays a big factor in whether the operators will be able to work well with the automation or not. Many of the studies suggested that the operators give direct feedback or be part of the process when it comes to interface design. The importance of training the operators was also

brought up showing that new machines require new knowledge in order to adapt to the new technology.

From this discussion, there are a few things that should be implemented by the companies in this industry. For example, they should first have a comprehensive training session with the operators to make sure they know how the machines work. They should also have a plan of action that they can rely on in case of incidents occurring when the engineers are not present. Another thing they can do is have regular, maybe monthly, meetings with some operators to ask if they have any suggestions to improve the overall design of the interface or see if anything is too complicated to work with. This would allow the operators to be more confident while working with the new automation. Lastly, they can make feedback loops so that the workers are encouraged to ask the engineers to improve the aspects that are holding them back. Overall, implementing these three improvements will enable better results for the company.

Limitations and Research Gaps

The current research does give many valuable insights, however, there are also a few limitations and research gaps that need to be addressed. A few of the studies that suggest design principles do not have enough practical evidence in the manufacturing industry to back up their findings (Neumann et al., 2021; Reiman et al., 2021). Field studies in actual production settings are rare in these research papers. There seems to be more laboratory research or theory than practical applications. Having a practical application would show us how these ideas are implemented in the industry and whether they are worth the cost that the companies would have to pay. This research gap is important because smart manufacturing systems are complex and dynamic with constraints that cannot be replicated in laboratory environments.

Finally, most studies talk about human factors such as ergonomics, trust, and usability individually. Instead, they should talk about how these factors interact with each other. For example, a poor interface might impact the trust that an operator has on the machine. Another thing that should be researched more is if ergonomic design influences the cognitive capacity of the operators while they are learning about the new technology. This is important because these factors often have an impact on other aspects of the automation process, and they should be researched more.

Future Research

Though this research talked about many aspects, there are still some items that are missing. This can be covered in future research. First, it is important to learn more about scalable ergonomic assessment tools that the organizations can use. These tools should be able to be used without expertise in them as there would be a wider implementation of them. With these tools, companies should be able to gather feedback from assessments that allow them to improve their design of the system.

Next, it is essential to expand human factors research to elaborate on the sustainability of the process and the well-being of the human workers (Longo et al., 2020). The metrics of the

company should also be aligned with the smart manufacturing principles that were discussed in the results section. This requires developing frameworks that work with companies that already have successful manufacturing systems and adding automation to the framework.

Lastly, it would be helpful to pinpoint the interactions between the human factors that were previously identified. This can be done by performing more holistic research designs that would provide a better understanding of how to optimize system performance. For example, the company could examine how the interface design that they chose works with the operators and if a better understanding allows them to foster trust with the automation system. This may also allow for less of a cognitive load as not understanding how something works can cause more stress. From the previous discussion, two things that all companies should implement is continuous training and feedback loops.

CONCLUSION

The literature review concluded that advanced technological development and automation is not enough by itself if the operators do not know how to make proper use of the new technology. There needs to be a human centric design to accomplish the smart manufacturing goals (Mattsson et al., 2020). The key points discussed in this paper are cognitive workload, trust and mistrust, and interface usability in the systems. Most of the articles iterate the fact that automation should be used as a helper for the human operator instead of completely replacing them on the production floor. This would allow the operators to appropriately trust the machines that they are working with and make the process more efficient. Along with technical competency, a good automation system also requires good feedback loops and comprehensive training.

There are a few recommendations that manufacturing companies should keep in mind. First, they should conduct training for the operators that allows them to learn how to properly work with the automation even if they do not have a background in technology. Next, there should always be a human-centric design. This means that the operators will not be afraid of being replaced by the machines. Lastly, the user interface design should make sense to the operator. This can be done by getting direct feedback from the users and keeping them in the design process. If the companies can implement these three principles, then their smart manufacturing system will be efficient.

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